



# Linear Models VI

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Stat 141 | Fall 2026  
Monday, Week 6

1

## Announcements

- Final exam is officially scheduled
- Extra time for quiz 3 revisions (due 10/16)

2

## Goals for Today

- Case Study
- Practice problems (dplyr and linear models)

3

## Case Study

4

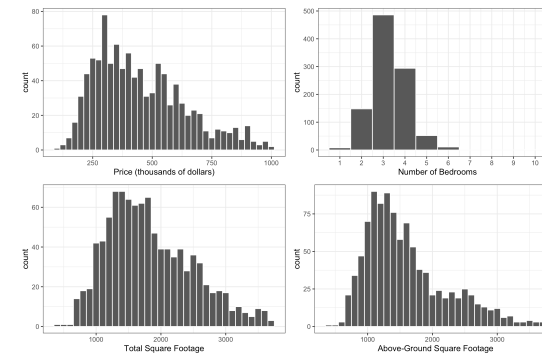
## Case Study: Home Prices in King County, WA

What's the relationship between a home's square footage and number of bedrooms on the home's sale price?

```
Rows: 1,000
Columns: 17
$ price      <dbl> 241.25, 262.50, 765.00, 430.00, 215.00, 675.00, 885.00, ...
$ bedrooms   <dbl> 3, 4, 4, 2, 3, 2, 4, 3, 3, 3, 2, 3, 4, 4, 2, 3, 5, 3, 3, ...
$ bathrooms  <dbl> 1.75, 2.00, 1.00, 2.25, 2.00, 1.75, 2.50, 1.50, 1.50, 2.00, ...
$ sqft_living <dbl> 1350, 1540, 2520, 1040, 1280, 2140, 2830, 1340, 1120, 1600, ...
$ sqft_lot   <dbl> 7588, 5110, 5500, 1516, 6994, 5000, 5000, 6000, 7000, 13000, ...
$ floors     <dbl> 1.0, 1.0, 1.5, 2.0, 1.0, 1.0, 2.0, 1.5, 1.0, 3.0, 1.0, 2.0, ...
$ waterfront <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, ...
$ view       <dbl> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, ...
$ condition  <dbl> 3, 3, 5, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 3, 3, 3, ...
$ grade      <dbl> 7, 7, 8, 8, 7, 7, 9, 9, 7, 9, 6, 9, 7, 8, 8, 7, 9, 6, 7, ...
$ sqft_above <dbl> 1350, 1540, 1820, 1040, 1280, 1000, 2830, 1340, 1120, 1300, ...
$ sqft_basement <dbl> 0, 0, 700, 0, 0, 1140, 0, 0, 0, 320, 480, 0, 890, 0, 0, ...
```

We'll focus on **four variables**: Sale price (*response*); and square footage, above-ground square footage, and number of bedrooms (*potential explanatory variables*).

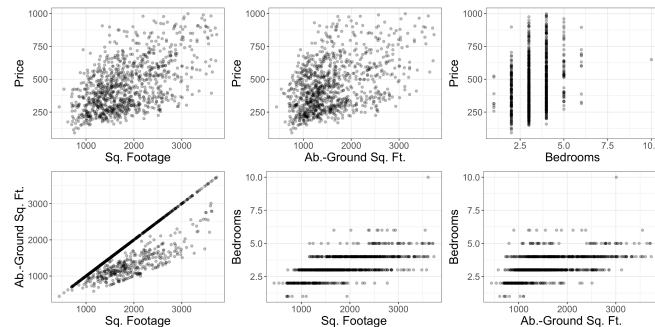
## Univariate Exploration



- Q: How would you describe the distribution of **price**?
- Q: What do you expect the relationship between **price** and **bedrooms** to be like?

02:00

## Bivariate Exploration, Multicollinearity



- Q: Which explanatory variables would be appropriate to include in our model?
- Q: Do any pairs of explanatory variables suffer from multicollinearity?

03:00

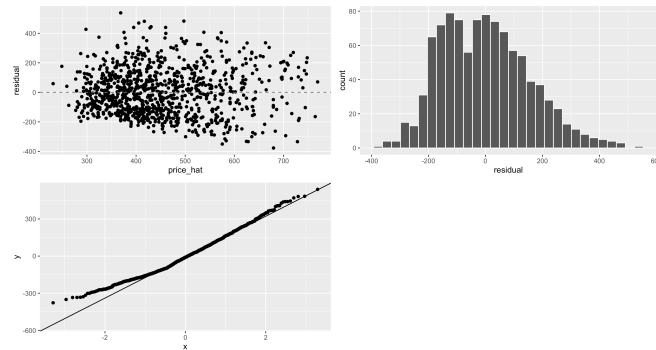
## Model Estimation

```
# A tibble: 3 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept  197.         19.4        10.1     0       159.    235.
2 sqft_living 0.177        0.01        18.4     0        0.158  0.196
3 bedrooms  -21.6         7.27        -2.97  0.003   -35.9   -7.34
```

- Q: What is the equation, including coefficient values, for the estimated model?
- Q: How should we interpret the **intercept** coefficient in our model?
- Q: How should we interpret the coefficient for **Bedrooms**? Does this match your expectation?

04:00

## LINE Assumptions



- Q: Does it seem like this model meets the four LINE assumptions?

03:00

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## Adjusted $R^2$

What's the adjusted  $R^2$  of our model?

```
1 get_regression_summaries(mod) %>% select("r_squared", "adj_r_squared")
# A tibble: 1 x 2
  r_squared adj_r_squared
<dbl>      <dbl>
1  0.314      0.313
```

Not super close to 1. Compare to a simpler model:

```
1 small_mod <- lm(price ~ bedrooms, data = houses)
2 get_regression_summaries(small_mod) %>% select("r_squared", "adj_r_squared")
# A tibble: 1 x 2
  r_squared adj_r_squared
<dbl>      <dbl>
1  0.082      0.081
```

Compare to a much bigger model with additional variables:

```
1 big_mod <- lm(price ~ bedrooms + bathrooms + sqft_living + sqft_above + sqft_lot + view + condition + yr_built,
2 data = houses)
3 get_regression_summaries(big_mod) %>% select("r_squared", "adj_r_squared")
# A tibble: 1 x 2
  r_squared adj_r_squared
<dbl>      <dbl>
1  0.434      0.429
```

- Q: Why are  $R^2$  and adjusted  $R^2$  different?

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## Practice problem: Data Wrangling Pseudo-Code

Looking at `pdxtrees` again:

```
# A tibble: 5 x 7
  Genus DBH Condition Tree_Height Crown_Width_NS Park Native
<chr> <dbl> <chr> <dbl> <dbl> <chr> <chr>
1 Quercus 3.3 Fair 16 14 Kenilworth Park No
2 Pseudotsuga 43.1 Fair 148 61 Kenilworth Park Yes
3 Thuja 14.1 Fair 42 21 Kenilworth Park Yes
4 Acer 36.5 Good 80 68 Woodstock Park No
5 Pinus 10.3 Fair 27 24 Sellwood Riverf... No
```

**Write pseudo-code to:** Make a contingency table of whether trees in Woodstock Park are “big guys” or “little guys” (one variable) and whether they are native species or not (second variable). Trees are considered “big guys” when DBH > 30 or Tree\_Height > 100 and “little guys” otherwise.

05:00

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## Practice problem: Data Wrangling Explaining Code

Now let's turn to the penguins data.

```
1 penguins %>%
2 filter(!is.na(sex),
3 island %in% c("Biscoe", "Dream")) %>%
4 count(species, island, sex) %>%
5 group_by(species, island) %>%
6 mutate(prop = n / sum(n)) %>%
7 arrange(species, island, sex)
```

**Explain what each line of code does in context, and sketch what the final data frame would look like.** Reminder that there are 3 different species in `penguins`: Adelies, Gentooes, and Chinstraps. It might be helpful to sketch what the data frame looks like at each intermediate step!

07:00

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## Vote: Another similar linear models problem, or more difficult linear models problem?

### Practice problem: More Linear Regression

In HW 4, we explored the results of the **Anchoring Effect Experiment** conducted during Lab 2. Our hypothesis is that the number a student saw on the paper slip (the **treatment**, either 82 or 531) affected the student's guess for the number of dog breeds in the world (the response variable).

We can assess the relationship between the guessed number of dog breeds and the number received on the paper slip by running a regression. Below is a summary of this regression, where **treatment** is treated as a **categorical** explanatory variable.

```
# A tibble: 2 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept  410.      188.      2.18  0.032    36.9   783.
2 treatment: 531  407.      259.      1.57  0.12   -108.   922.
```

- **Q3:** Suppose I had tested the effect of three different treatments: 82, 531, and 900, instead of just 82 and 531. What would be different about the regression model and the results table?
- **Q4:** Suppose **treatment** was actually a **quantitative** variable, with each student seeing a different number in [0, 200]. Suppose that then our regression model produced estimates  $\hat{\beta}_0 = 100$  and  $\hat{\beta}_1 = 4$ . How should we interpret  $\hat{\beta}_0$  and  $\hat{\beta}_1$  now?

05:00

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### Practice problem: More Linear Regression (skip?)

In HW 4, we explored the results of the **Anchoring Effect Experiment** conducted during Lab 2. Our hypothesis is that the number a student saw on the paper slip (the **treatment**, either 82 or 531) affected the student's guess for the number of dog breeds in the world (the response variable).

We can assess the relationship between the guessed number of dog breeds and the number received on the paper slip by running a regression. Below is a summary of this regression, where **treatment** is treated as a **categorical** explanatory variable.

```
# A tibble: 2 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept  410.      188.      2.18  0.032    36.9   783.
2 treatment: 531  407.      259.      1.57  0.12   -108.   922.
```

- **Q1:** Write the equation for the **estimated regression line**.
- **Q2:** How should we interpret the “intercept” and “**treatment: High**” terms, given the context of the experiment?

03:00

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### Practice problem: Linear Regression (extra difficult 🐧)

Still hanging with the penguins... Take a look at the following regression table:

```
# A tibble: 5 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept -9.83      4.83     -2.04  0.043   -19.3   -0.326
2 sex: male  5.89      2.60      2.27  0.024    0.779   11.0
3 body_mass 0          0.001     -0.182 0.856   -0.002    0.001
4 flipper_len 0.266    0.033      7.93  0        0.2      0.332
5 sex: male:body_mass -0.001    0.001     -1.41  0.161   -0.002    0
```

- **Q1:** How should we interpret **intercept**?
- **Q2:** How should we interpret **body\_mass**?
- **Q3:** How should we interpret **flipper\_len**?

08:00

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